

Rockchip LVDS Interface Developer Guide

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Preface

This article mainly introduces various configurations and debugging verification methods of the LVDS display interface on the Rockchip platform.

Product Version

Chipset	Kernel Version
RK3126	Linux 4.4
RK3128	Linux 4.4
RK3288	Linux 4.4
RK3326/PX30	Linux 4.19 and later versions
RK3368	Linux 4.19 and later versions
RK356X	Linux 4.19 and later versions

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Change Description
V1.0.0	Chaoyi Chen	2023-11-16	Initial version
V1.1.0	Chaoyi Chen	2023-12-25	Supplementary LVDS use cases, added illustrations
V1.2.0	Chaoyi Chen	2024-7-19	Supplementary LVDS electrical characteristics and RK356X common-mode/differential-mode voltage configurations

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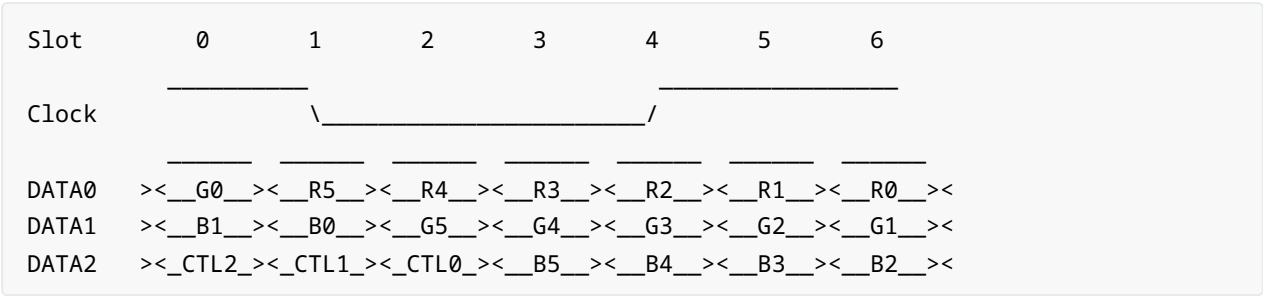
1. Basic Concepts

LVDS (Low Voltage Differential Signaling) is a low-voltage differential signal transmission technology used to transmit signals in high-speed serial data communication. It is widely applied in fields such as display, image transmission, and data communication.

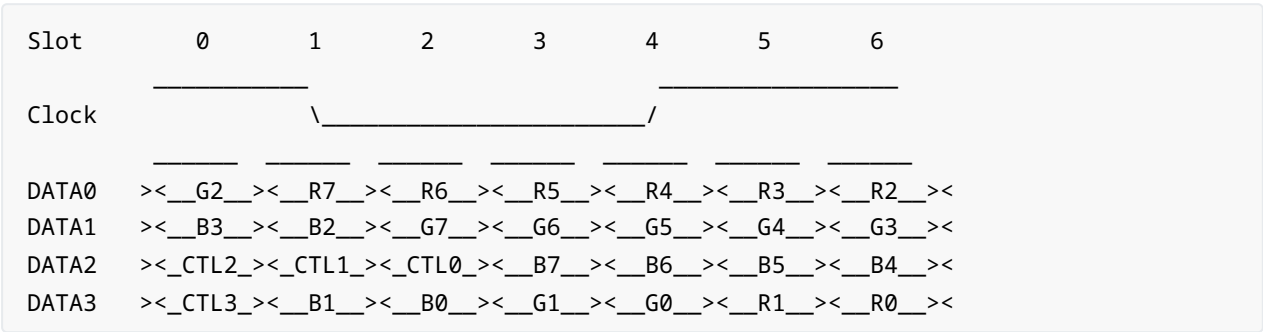
LVDS is a physical layer signal transmission technology that defines electrical characteristics and interface standards. The VESA and JEIDA organizations stipulate the transmission methods of LVDS on the data link. According to the amount of transmitted data and the transmission sequence, it can be divided into JEIDA-18, JEIDA-24, VESA-24, etc.

For LVDS, 7 bits of data can be transmitted in one clock cycle. The following briefly introduces the timings of various common LVDS formats.

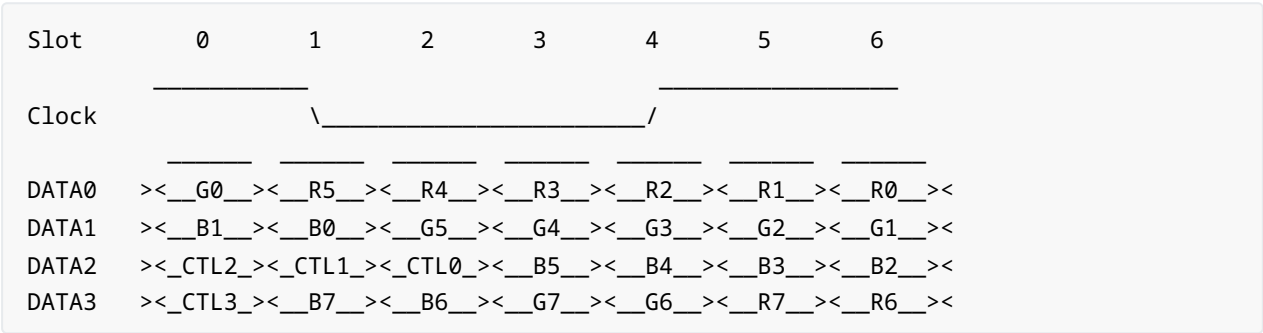
1. JEIDA-18



2. JEIDA-24



3. VESA-24

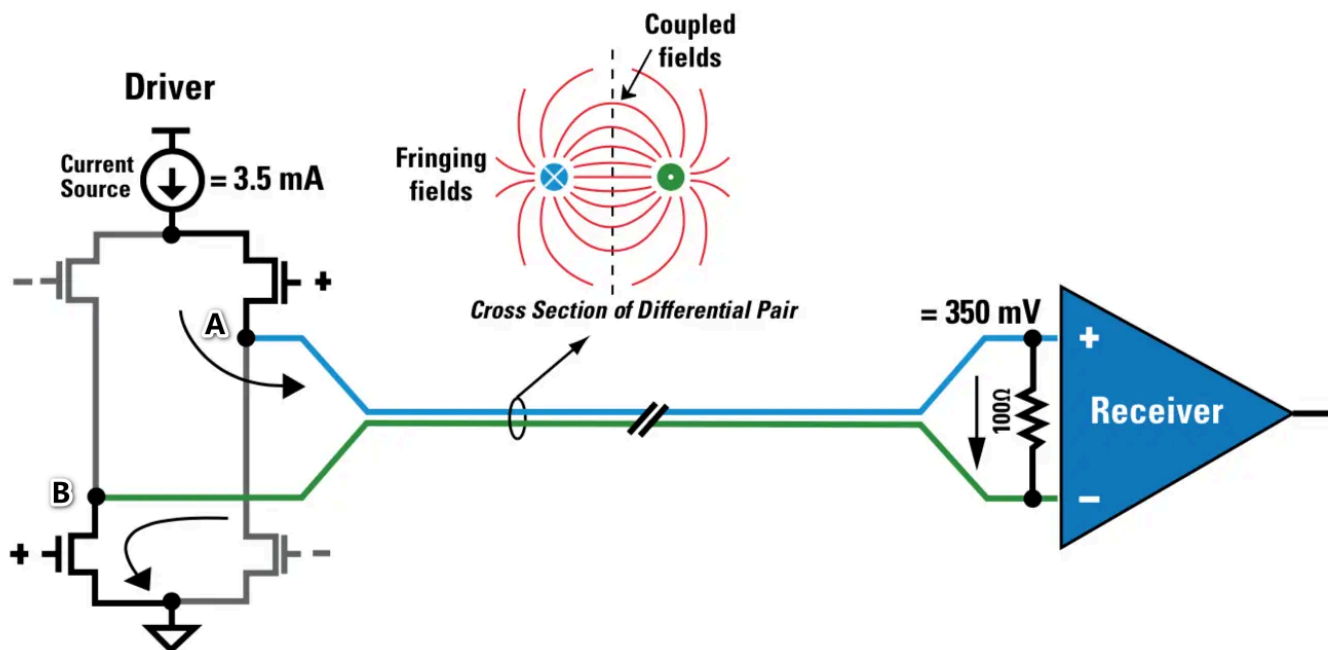


Among them, CTL2 is the DE enable signal, CTL1 is the VSYNC field sync signal, and CTL0 is the HSYNC line sync signal. CTL3 is a reserved signal for additional purposes and can generally be ignored.

In the panel node of DTS, the LVDS format to be adopted can be specified through the bus-format attribute, and the corresponding relationships are as follows:

- MEDIA_BUS_FMT_RGB666_1X7X3_SPWG : jeida-18
- MEDIA_BUS_FMT_RGB888_1X7X4_JEIDA : jeida -24
- MEDIA_BUS_FMT_RGB888_1X7X4_SPWG : vesa-24

2. Electrical Characteristics



The typical implementation circuit of the LVDS driver is shown in the figure above. Essentially, it is a differential amplifier circuit driven by a constant current source.

First, consider the DC output characteristics. For an ideal differential amplifier circuit, the common-mode voltage V_{cm} can be expressed in the following form:

$$V_{cm} = \frac{U_a + U_b}{2}$$

where U_a is the DC voltage of A to ground in the figure, and U_b is the DC voltage of B to ground in the figure.

For an ideal differential amplifier circuit, it is considered that $U_a = U_b$, that is, there is no voltage difference across the 100 Ω resistor at the receiving end, showing the characteristic of DC isolation.

Then consider the AC output characteristics. The small AC signal u_{ab} generated at both ends of AB produces a voltage drop on both sides of the 100 Ω resistor at the receiving end, showing the characteristic of AC passing.

A common requirement is to adjust the driving strength of LVDS. Taking enhancing the driving strength as an example, there are various implementation methods, such as increasing the current of the constant current source and increasing the load resistance.

The above analysis is from the perspective of circuit analysis. In practical applications, some LVDS PHYs provide additional configurations to complete these operations. For example, the driving strength of LVDS can be increased by increasing the differential voltage U_{ab} (the higher the voltage difference, the greater the current).

In practical applications, the impact of the transmission line also needs to be considered. Optimizing the capacitance and resistance on the differential line (making them smaller) can also enhance the driving strength of LVDS to a certain extent.

3. Platform Support

Platform	Support LVDS0	Support LVDS1
RK3568	Y	Y
RK3567	Y	Y
RK3566	Y	N
RK3562	Y	N
RK3368	Y	N
RK3326 / PX30	Y	N
RK3288	Y	Y
RK3128	Y	N
RK3126	Y	N

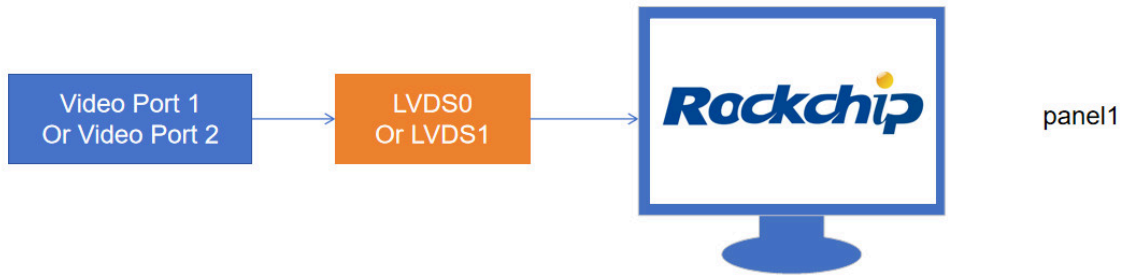
4. Application Scenario Overview

According to different usage scenarios, the usage of LVDS can be summarized in the following table:

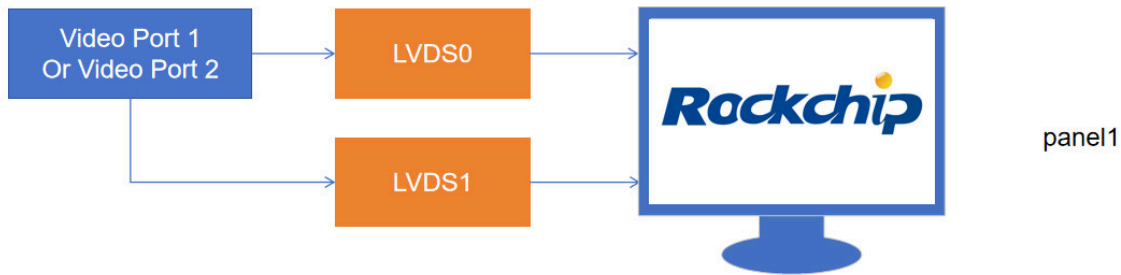
No.	Number of panels	LVDS Interface Connection	Video Port Output Allocation	Maximum Resolution	Description	Reference Configuration
1	1 panel	LVDS0 or LVDS1	VP1 -> LVDS0/1 or VP2 -> LVDS0/1	1280x800	Single-channel LVDS panel	rk3567-evb2-lp4x-v10-single-channel-lvds.dts and rk3568-evb1-ddr4-v10-single-channel-lvds.dts
2	1 panel	LVDS0 and LVDS1	VP1 -> LVDS0, VP1-> LVDS1 or VP2 -> LVDS0, VP2-> LVDS1	1920x1080	Dual LVDS, dual-channel LVDS	rk3567-evb2-lp4x-v10-dual-channel-lvds.dts
3	2 panel	LVDS0 and LVDS1	VP1 -> LVDS0, VP2 -> LVDS1 or VP2 -> LVDS0, VP1 -> LVDS1	1280x800 per panel	Two VPs are respectively connected to a single-channel LVDS panel	rk3567-evb2-lp4x-v10-two-vp-two-separate-single-channel-lvds.dts and rk3568-evb1-ddr4-v10-two-vp-two-separate-single-channel-lvds.dts
4	2 panels	LVDS0 and LVDS1	VP1 -> LVDS0, VP1-> LVDS0 or VP2 -> LVDS0, VP2-> LVDS0	1280x800 per panel	One VP is connected to two single-channel LVDS panels, and each panel displays half of the content in the VP	rk3567-evb2-lp4x-v10-one-vp-two-single-channel-lvds.dts and rk3568-evb1-ddr4-v10-one-vp-two-single-channel-lvds.dts
5	2 panels	LVDS0 and LVDS1	VP1 -> LVDS0, VP1-> LVDS1 or VP2 -> LVDS0, VP2-> LVDS1	1280x800 per panel	One VP is connected to two single-channel LVDS panels, and each panel displays exactly the same content (connector mirror)	See the relevant section in this article

The actual effects of these 5 scenarios are shown in the following figure:

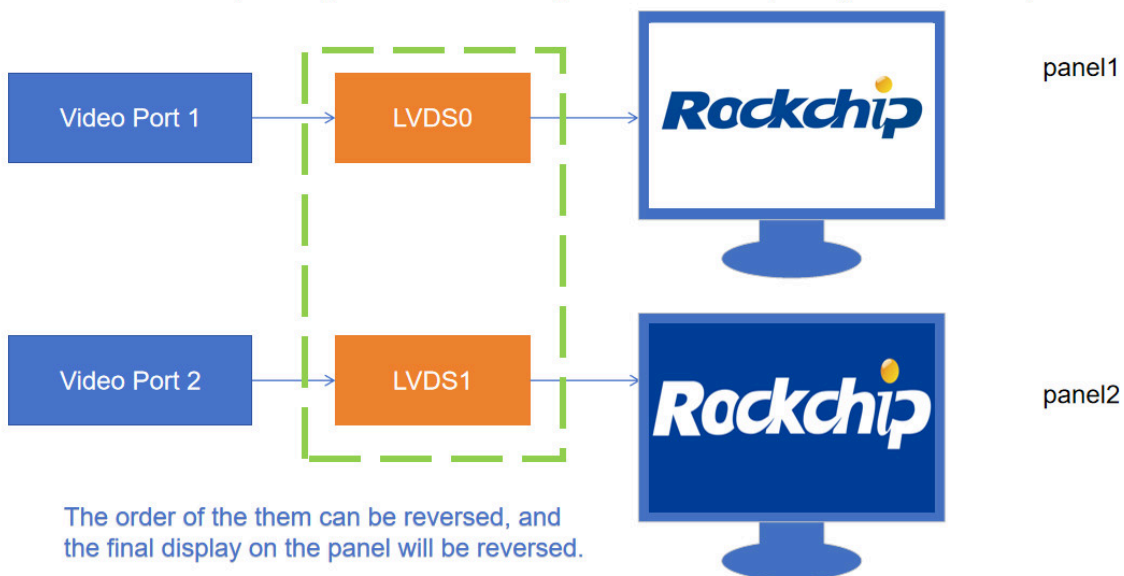
1. Single-Channel LVDS panel



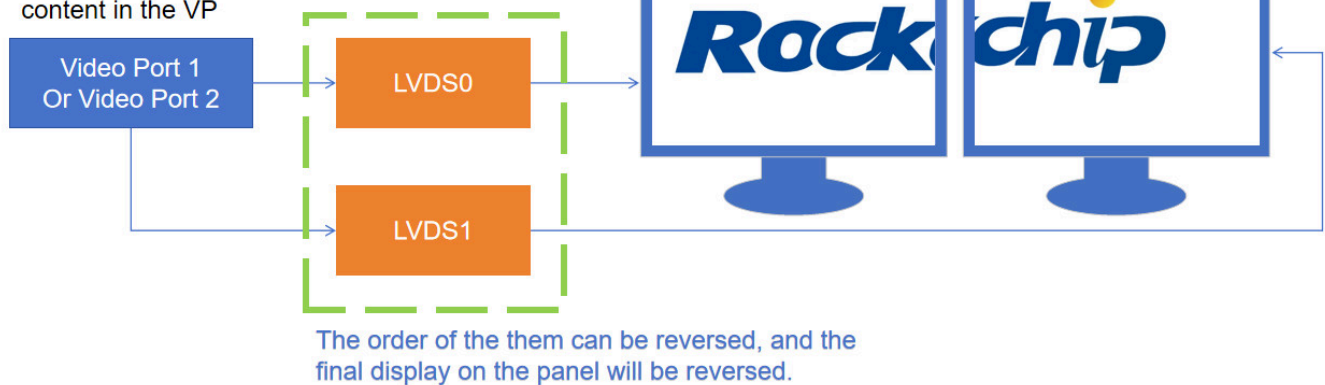
2. Dual LVDS (Dual-Channel LVDS)



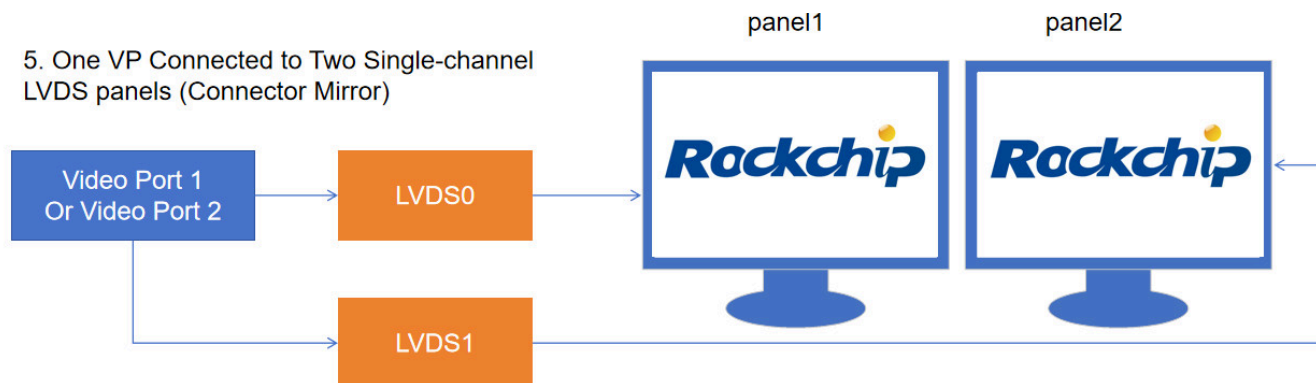
3. Two VPs are respectively connected to a single-channel LVDS panel (in reverse order)



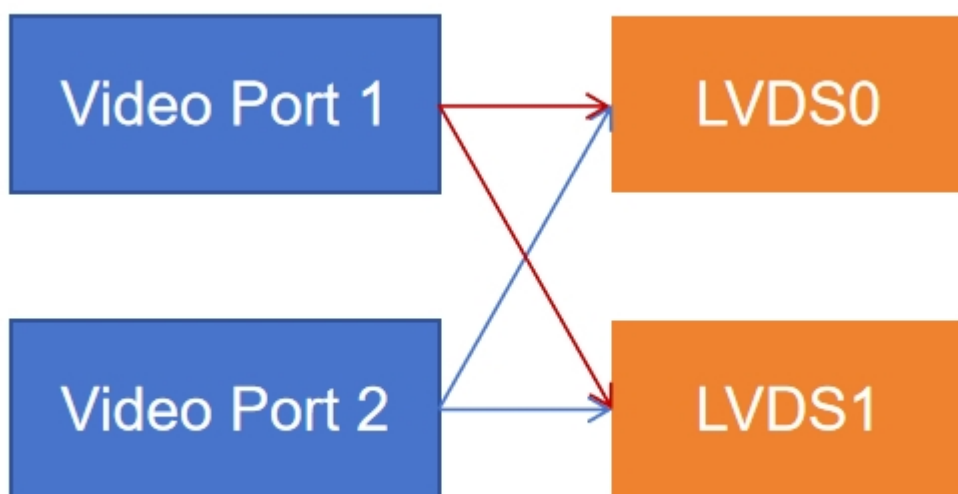
4. One VP is connected to two single-channel LVDS panels, and each panel shows half of the content in the VP



5. One VP Connected to Two Single-channel LVDS panels (Connector Mirror)



For RK356X, the allocation relationship between the Video Port and LVDS is shown in the following figure:



5. LVDS Configuration Examples for Different Scenarios

In this section, configuration examples of different LVDS will be introduced. Although there are various combinations of LVDS outputs, in fact, only the following two points need to be considered:

- Whether it is dual-channel LVDS, that is, Dual LVDS.
- Which VP the actual LVDS corresponds to.

For Dual LVDS, two LVDS nodes need to be enabled, and the `dual-channel` attribute needs to be added to the LVDS0 node to indicate that Dual LVDS is used.

```
&lvds0 {  
    status = "okay";  
    dual-channel;  
};
```

In addition, it is necessary to confirm which VP the LVDS corresponds to. For Dual LVDS, both LVDS0 and LVDS1 use VP1 or both use VP2 as the output. For the remaining cases, the configuration is carried out according to the specific scenario:

```
/*
 * Example 1: Two VPs are respectively connected to a single-channel LVDS panel
 * VP1 -> LVDS0, VP2 -> LVDS1
 */
&lvds0_in_vp1 {
    status = "okay";
};

&lvds1_in_vp2 {
    status = "okay";
};

/*
 * Example 2: Two VPs are respectively connected to a single-channel LVDS panel (in reverse
 order)
 * VP2 -> LVDS0, VP1 -> LVDS1
 */
&lvds0_in_vp2 {
    status = "okay";
};

&lvds1_in_vp1 {
    status = "okay";
};
```

5.1 Single-channel LVDS Configuration

When configuring single-channel LVDS, the following parameters need to be confirmed:

- The panel timing, output format, corresponding Enable pin, and PWM backlight pin
- Whether to output from LVDS0 or LVDS1
- Whether to output from VP1 or VP2 to LVDS

For single-channel LVDS on RK356X, there are 4 allocation relationships between LVDS and VP, as shown in the following figure:

1.



2.



3.



4.



Taking VP1 outputting LVDS0 as an example, the following briefly introduces the DTS configuration process:

1. Set the panel node, configure the panel timing, output format, Enable, and backlight pins, and associate the panel with the LVDS node:

```
panel {  
    compatible = "simple-panel";  
    backlight = <&backlight>;  
    power-supply = <&vcc3v3_lcd0_n>;  
    enable-delay-ms = <20>;  
    prepare-delay-ms = <20>;  
    unprepare-delay-ms = <20>;  
    disable-delay-ms = <20>;  
    bus-format = <MEDIA_BUS_FMT_RGB666_1X7X3_SPWG>;  
    width-mm = <217>;  
    height-mm = <136>;  
  
    display-timings {  
        native-mode = <&timing0>;  
        timing0: timing0 {  
            clock-frequency = <67000000>;  
            hactive = <800>;  
        };  
    };  
};
```

```

        vactive = <1280>;
        hback-porch = <60>;
        hfront-porch = <60>;
        vback-porch = <4>;
        vfront-porch = <2>;
        hsync-len = <8>;
        vsync-len = <8>;
        hsync-active = <0>;
        vsync-active = <0>;
        de-active = <0>;
        pixelclk-active = <0>;
    };
};

ports {
    #address-cells = <1>;
    #size-cells = <0>;
    port@0 {
        reg = <0>;
        panel_in_lvds0: endpoint {
            remote-endpoint = <&lvds0_out_panel>;
        };
    };
};
};

```

2. Enable the corresponding LVDS node and associate it with the panel node:

```

&lvds0 {
    status = "okay";
    ports {
        port@1 {
            reg = <1>;
            lvds0_out_panel: endpoint {
                remote-endpoint = <&panel_in_lvds0>;
            };
        };
    };
};

```

3. Specify on which VP the LVDS is output:

```

&lvds0_in_vp1 {
    status = "okay";
};

```

5.2 Dual LVDS (Dual-channel LVDS) Configuration

The configuration in this section is applicable to RK356X.

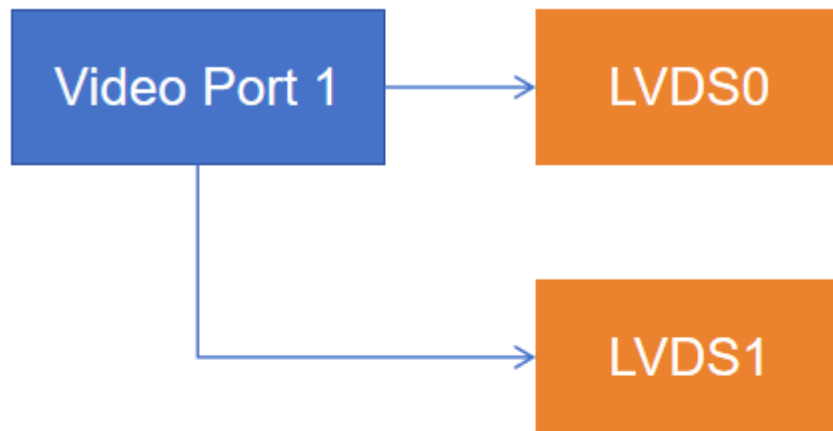
1. Add the `dual-channel` attribute to the `lvds` node of LVDS0 to indicate that Dual LVDS is used.

```
&lvds0 {
    status = "okay";
    dual-channel;

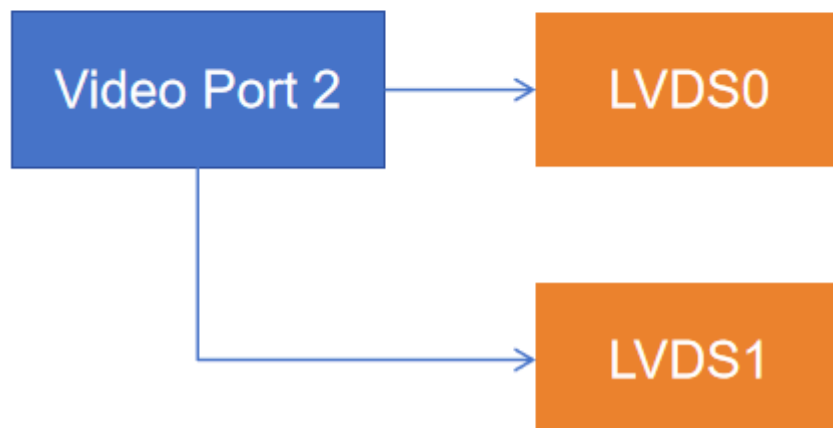
    ports {
        port@1 {
            reg = <1>;
            lvds0_out_panel: endpoint {
                remote-endpoint = <&panel_in_lvds0>;
            };
        };
    };
};
```

2. Specify the allocation method of the Video Port. For Dual LVDS on RK356X, it can be selected to output on either VP1 or VP2. The following figure shows all possible allocation methods.

1.



2.



The following example shows the output using VP1, that is, Method 1 in the figure:

```
&lvds0_in_vp1 {
    status = "okay";
};

&lvds1_in_vp1 {
    status = "okay";
};

&lvds1_in_vp2 {
    status = "disabled";
};
```

3. Specify the output order of the LVDS channels:

```
panel {
    /* ... Relevant panel parameters are filled in here */

    ports {
        #address-cells = <1>;
        #size-cells = <0>;
        port@0 {
            reg = <0>;
            dual-lvds-odd-pixels; /* Output odd pixels */
            panel_in_lvds0: endpoint {
                remote-endpoint = <&lvds0_out_panel>;
            };
        };
        port@1 {
            reg = <1>;
            dual-lvds-even-pixels; /* Output even pixels */
            panel_in_lvds1: endpoint {
                remote-endpoint = <&lvds1_out_panel>;
            };
        };
    };
};
```

In this example, LVDS0 outputs `dual-lvds-odd-pixels`, that is, odd pixels; LVDS1 outputs `dual-lvds-even-pixels`, that is, even pixels. In practice, the order of the them can be swapped as needed.

5.3 Dual LVDS (Dual-channel LVDS) Configuration Applicable to RK3288

This section is applicable to RK3288.

1. Add the `dual-channel` attribute to the `lvds` node to indicate that Dual LVDS is used.

```
&lvds {
    status = "okay";
    dual-channel;

    ports {
        port@1 {
            reg = <1>;
            lvds0_out_panel: endpoint {
                remote-endpoint = <&panel_in_lvds0>;
            };
        };
    };
};
```

2. Specify on which VOP the LVDS is output, which can be vopb or vopl.

```
/* Output on vopb */
&lvds_in_vopb {
    status = "okay";
};

/* Or, output on vopl */
&lvds_in_vopl {
    status = "okay";
}
```

3. Specify the output order of the LVDS channels. For RK3288, the `rockchip,data-swap` attribute can be added to the `lvds` node to swap the output order:

```
&lvds {
    rockchip,data-swap;
};
```

5.4 Configuration for Two VPs Each Connected to an Independent Single-channel LVDS Panel

In this case, only need to configure each single-channel LVDS according to the single-channel LVDS configuration.

For this case, do not set the `dual-channel` attribute, otherwise it will be recognized as Dual LVDS

In practical use, the allocation method of the Video Port needs to be specified. There are 2 possible allocation methods, as shown in the following figure:

1.



2.



The following DTS examples respectively correspond to Method 1 and Method 2 in the figure:

```
/*
 * VP1 -> LVDS0, VP2 -> LVDS1
 */
&lvds0_in_vp1 {
    status = "okay";
};

&lvds1_in_vp2 {
    status = "okay";
};

/*
 * Or, VP2 -> LVDS0, VP1 -> LVDS1
 */
&lvds0_in_vp2 {
    status = "okay";
};

&lvds1_in_vp1 {
    status = "okay";
};
```

5.5 Configuration for One VP Connected to Two Single-channel LVDS Panels

In this scenario, one VP is connected to two single-channel LVDS panels, and each panel shows half of the content in the VP.

It is worth noting that in this case, it is currently implemented as an extension of Dual LVDS. Therefore, its configuration method is similar to that of Dual LVDS:

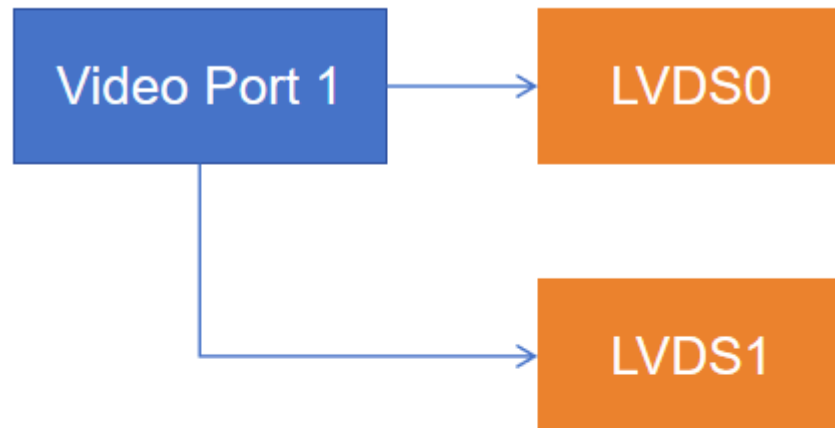
1. Add the `dual-channel` attribute to the `lvds` node of LVDS0.

```
&lvds0 {
    status = "okay";
    dual-channel;

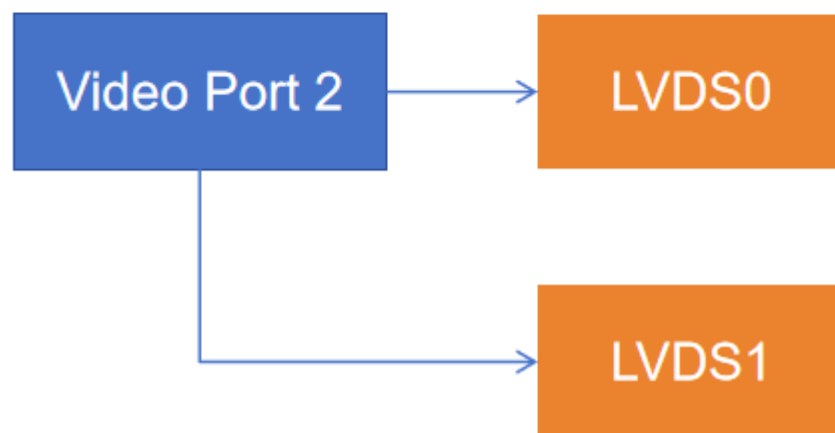
    ports {
        port@1 {
            reg = <1>;
            lvds0_out_panel: endpoint {
                remote-endpoint = <&panel_in_lvds0>;
            };
        };
    };
};
```

2. Specify the allocation method of the Video Port. For the scenario of one VP connected to two single-channel LVDS panels on RK356X, it can be selected to output on either VP1 or VP2. The possible allocation methods are shown in the following figure.

1.



2.



The following example shows the output using VP1, that is, Method 1 in the figure:

```
&lvds0_in_vp1 {  
    status = "okay";  
};  
  
&lvds1_in_vp1 {  
    status = "okay";  
};  
  
&lvds1_in_vp2 {  
    status = "disabled";  
};
```

3. Specify the output order of the LVDS channels:

```
panel {  
    /*... Relevant panel parameters are filled in here */  
  
    ports {  
        #address-cells = <1>;  
        #size-cells = <0>;  
        port@0 {
```

```

        reg = <0>;
        dual-lvds-left-pixels; /* Output the left-half content */
        panel_in_lvds0: endpoint {
            remote-endpoint = <&lvds0_out_panel>;
        };
    };
    port@1 {
        reg = <1>;
        dual-lvds-right-pixels; /* Output the right-half content */
        panel_in_lvds1: endpoint {
            remote-endpoint = <&lvds1_out_panel>;
        };
    };
};
};
};

```

In the current implementation, we have additionally defined the `dual-lvds-left-pixels` and `dual-lvds-right-pixels` attributes to distinguish the output order. It can be changed as needed in practical use.

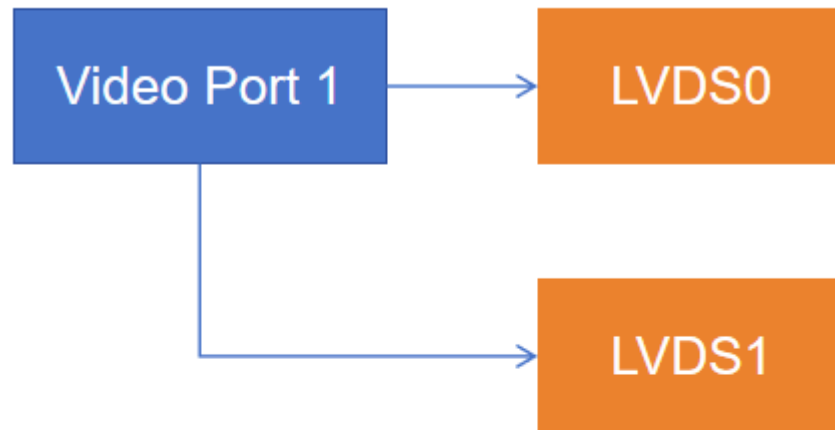
5.6 Configuration for One VP Connected to Two Single-channel LVDS panels (Connector Mirror)

In this scenario, one VP is connected to two single-channel LVDS panels, and each panel has the same timing parameters and displays exactly the same content.

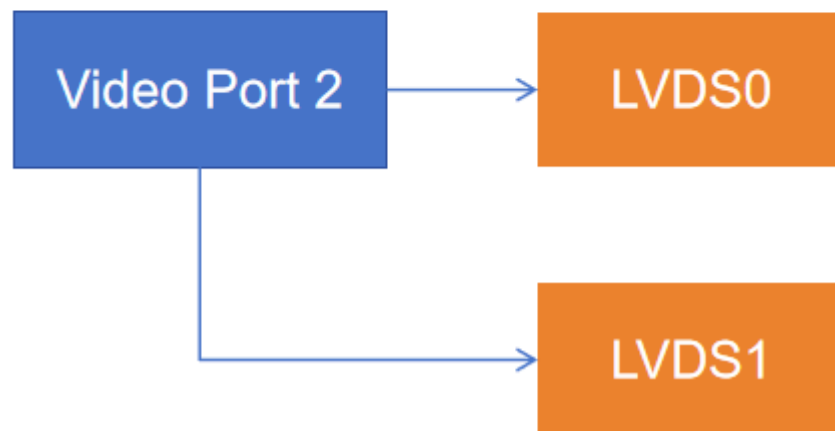
For the configuration of each panel's timing, it can refer to the single-channel LVDS configuration.

In terms of the allocation method of the Video Port and LVDS, it can be specified to output from either VP1 or VP2:

1.



2.



```
/* Output using VP1 */
&lvds0_in_vp1 {
    status = "okay";
};

&lvds1_in_vp1 {
    status = "okay";
};

/* Or, output using VP2 */
&lvds0_in_vp2 {
    status = "okay";
};

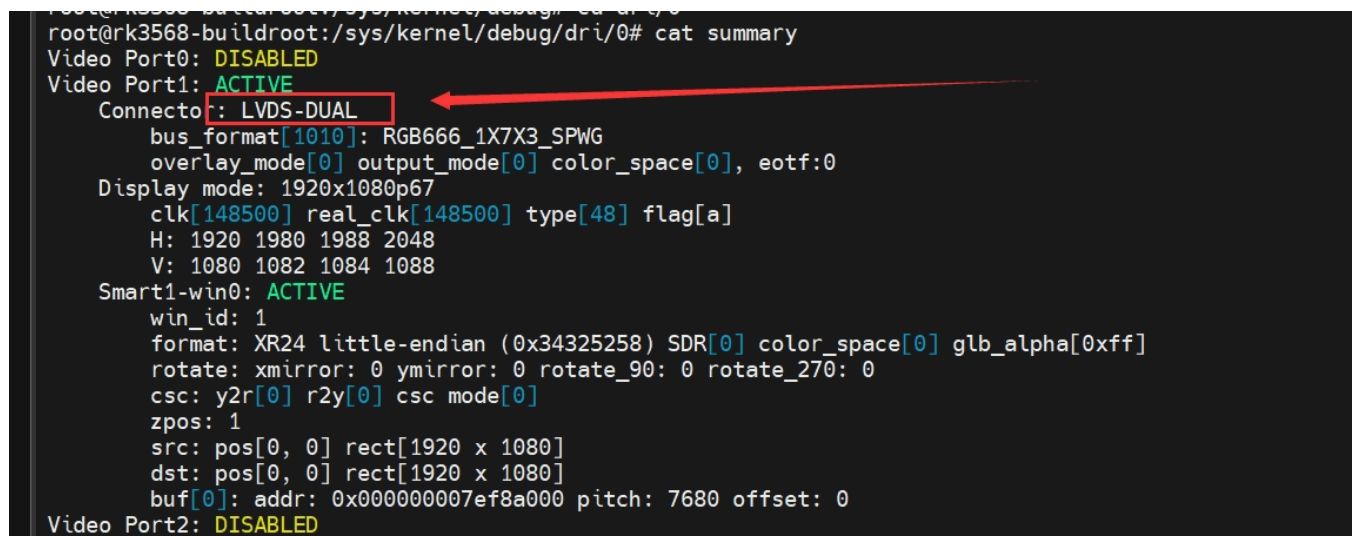
&lvds1_in_vp2 {
    status = "okay";
};
```

6. LVDS Debugging Method

With the help of the summary node of VOP, the current LVDS Connector information can be viewed:

```
cat /sys/kernel/debug/dri/0/summary
```

When Dual LVDS is enabled and the driver has been successfully loaded, the Connector node named **LVDS-DUAL** will be displayed:

A terminal window screenshot showing the output of the command 'cat /sys/kernel/debug/dri/0/summary'. The output lists various video port and connector settings. A red box highlights the 'Connector: LVDS-DUAL' line, with a red arrow pointing to it from the right. Other visible lines include 'Video Port0: DISABLED', 'Video Port1: ACTIVE', 'bus_format[1010]: RGB666_1X7X3_SPWG', 'Display mode: 1920x1080p67', 'Smart1-win0: ACTIVE', and 'Video Port2: DISABLED'.

```
root@rk3568-buildroot:/sys/kernel/debug/dri/0# cat summary
Video Port0: DISABLED
Video Port1: ACTIVE
Connector: LVDS-DUAL
  bus_format[1010]: RGB666_1X7X3_SPWG
  overlay_mode[0] output_mode[0] color_space[0], eotf:0
Display mode: 1920x1080p67
  clk[148500] real_clk[148500] type[48] flag[a]
  H: 1920 1980 1988 2048
  V: 1080 1082 1084 1088
Smart1-win0: ACTIVE
  win_id: 1
  format: XR24 little-endian (0x34325258) SDR[0] color_space[0] glb_alpha[0xff]
  rotate: xmirror: 0 ymirror: 0 rotate_90: 0 rotate_270: 0
  csc: y2r[0] r2y[0] csc mode[0]
  zpos: 1
  src: pos[0, 0] rect[1920 x 1080]
  dst: pos[0, 0] rect[1920 x 1080]
  buf[0]: addr: 0x000000007ef8a000 pitch: 7680 offset: 0
Video Port2: DISABLED
```

Note that for the scenario where one VP is connected to two single-channel LVDS panels and each panel displays half of the content in the VP, it will also be displayed as LVDS-DUAL.

When two single-channel LVDS panels are used, different LVDS Connector nodes can be seen:

```

Video Port0: DISABLED
Video Port1: ACTIVE
  Connector: LVDS-2
    bus_format[1010]: RGB666_1X7X3_SPWG
    overlay_mode[0] output_mode[0] color_space[0], eotf:0
  Display mode: 800x1280p56
    clk[67000] real_clk[67000] type[48] flag[a]
    H: 800 860 868 928
    V: 1280 1282 1290 1294
  Cluster1-win0: ACTIVE
    win_id: 6
    format: AB24 little-endian (0x34324241)[AFBC] SDR[0] color_space[0] glb_alpha[0xff]
    rotate: xmirror: 0 ymirror: 0 rotate_90: 0 rotate_270: 0
    csc: y2r[0] r2y[0] csc mode[0]
    zpos: 0
    src: pos[0, 0] rect[800 x 1280]
    dst: pos[0, 0] rect[800 x 1280]
    buf[0]: addr: 0x0000000007f350000 pitch: 3200 offset: 0
Video Port2: ACTIVE
  Connector: LVDS-1
    bus_format[1010]: RGB666_1X7X3_SPWG
    overlay_mode[0] output_mode[0] color_space[0], eotf:0
  Display mode: 800x1280p56
    clk[67000] real_clk[66667] type[48] flag[a]
    H: 800 860 868 928
    V: 1280 1282 1290 1294
  Esmart1-win0: ACTIVE
    win_id: 2
    format: AB24 little-endian (0x34324241) SDR[0] color_space[0] glb_alpha[0xff]
    rotate: xmirror: 0 ymirror: 0 rotate_90: 0 rotate_270: 0
    csc: y2r[0] r2y[0] csc mode[0]
    zpos: 0
    src: pos[0, 0] rect[800 x 1280]
    dst: pos[0, 0] rect[800 x 1280]
    buf[0]: addr: 0x0000000007e788000 pitch: 3200 offset: 0

```

At this time, which VP the LVDS Connector is allocated to depends on the specific configuration.

7. FAQ

7.1 No Output on LVDS1 Channel

First, check the output information of VOP summary to ensure that LVDS1 has been correctly enabled. If there is still no output, try to update the Loader.

7.2 Blurred/Jagged Content Output by Dual LVDS

It is possible that the outputs of LVDS0 and LVDS1 are reversed. You can try to swap the positions of the `dual-lvds-odd-pixels` and `dual-lvds-even-pixels` attributes in the DTS configuration.

7.3 Display content is Black/White

Check the hardware connection and whether the outputs of the backlight pin and the Enable pin are correct.

7.4 Modify LVDS Common-mode/Differential-mode Voltage

For RK356X, modify the DTS attributes corresponding to the PHY:

Among them, `inno,lvds-vcom` is the common-mode voltage, and `inno,lvds-vod` is the differential-mode voltage.

The common-mode voltages of RK356X are as follows:

- 1000 mV
- 950 mV (default value)
- 900 mV
- 850 mV

The differential-mode voltages of RK356X are as follows:

- 400 mV
- 350 mV (default value)
- 300 mV
- 250 mV

```
&video_phy0 {  
    inno,lvds-vcom = <1000>; /* 1000mV */  
    inno,lvds-vod = <400>; /* 400mV */  
};
```